Reflectivity-Augmented LiDAR Scene Understanding for Robotic Perception

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***Abstract*—This work investigates whether a lightweight, range-aware transformation of LiDAR intensity can serve as a more useful semantic cue than raw intensity alone for robotic perception. Raw intensity is not a clean intrinsic surface property: it is entangled with range, incidence geometry, and sensor response. We adopt a deliberately simple pseudo-reflectivity proxy, $\hat{\rho}_i = I_i \cdot R_i$, where I_i is raw intensity and R_i is point-wise Euclidean range, and build a structured multi-stage evaluation on SemanticKITTI sequence 00 spanning a 15-window framework over short, medium, and long temporal scales, multiple proxy variants, semantic separability analysis, failure-case characterization, and a lightweight downstream classification study. The direct proxy $I \cdot R$ produces measurable semantic gains but exhibits high variability and explicit failure modes, while $\log(1 + I \cdot R)$ yields more consistent improvements. On a 600-frame contiguous run, pseudo-reflectivity improves frame-level Fisher separability in 400 of 600 frames (mean $\Delta = +0.0436$, $+14.45\%$ relative). A logistic-regression downstream test using $[I, I \cdot R]$ over $[I]$ yields a $+3.11$-point overall gain, concentrated on road ($+11.6\%$) and sidewalk ($+7.6\%$). The work establishes reflectivity-aware augmentation as a controlled, interpretable signal enhancement whose strength is scene- and proxy-dependent.**

I. Introduction

LiDAR sensors provide robotic systems with sparse but geometrically meaningful measurements of the environment. Each point carries a spatial coordinate triple and a recorded return strength, commonly referred to as *intensity*. While intensity is informative, it is not equivalent to surface reflectivity. Under a simplified return-strength model, intensity is jointly determined by surface reflectivity, incidence angle, range, and sensor-specific response. Treating raw intensity as a stable surface descriptor therefore conflates intrinsic material behavior with measurement geometry.

The midterm phase of this project [1], [2] established that a deliberately simple range-aware transformation, $\hat{\rho} = I \cdot R$, reshapes the LiDAR signal in a way that improves semantic interpretability on a single SemanticKITTI frame and over short motion windows. The final project extends this into a structured multi-stage evaluation that asks: *Does the pseudo-reflectivity proxy generalize across multiple temporal segments? Are there proxy formulations more stable than $I \cdot R$? Where does the method fail, and why? And does it provide measurable utility in a lightweight downstream classification setting?*

This work focuses on signal-level analysis: characterizing how a controlled, range-aware feature transformation reshapes LiDAR intensity, quantifying its semantic impact across multiple windows, and identifying the conditions under which it helps. The contribution is a structured evaluation methodology applied to a simple, physically motivated idea, with reproducible artifacts and explicit per-stage outputs.

II. Concepts and Mathematical Model

Each LiDAR point is represented as

$$p_i = (x_i, y_i, z_i, I_i), \tag{1}$$

where (x_i, y_i, z_i) are Cartesian coordinates in the sensor frame and I_i is recorded intensity. For each point, Euclidean range is computed as

$$R_i = \sqrt{x_i^2 + y_i^2 + z_i^2}. \tag{2}$$

A simplified LiDAR return-strength model is

$$I_i \propto \frac{\rho_i \cos(\alpha_i)}{R_i^2}, \tag{3}$$

where ρ_i is intrinsic surface reflectivity and α_i is the incidence angle between the laser ray and the surface normal. Equation (3) makes explicit that raw intensity entangles intrinsic surface behavior, range attenuation, and incidence geometry. Recovering ρ_i exactly would require known incidence angles, sensor calibration, and atmospheric correction terms that are not available within the scope of this work.

A. Pseudo-reflectivity Proxy

Instead of attempting full reflectivity recovery, this work uses a lightweight, range-aware proxy:

$$\hat{\rho}_i = I_i \cdot R_i. \tag{4}$$

This is not calibrated reflectivity. It is a heuristic, range-aware transformation chosen to expose more semantic structure than raw intensity at minimal computational cost. The choice of R^1 rather than R^2 is empirical: as shown in Section V, the more aggressive $I \cdot R^2$ variant exhibits severe distribution explosion and is dominated by far-range points, while $I \cdot R$ remains stable enough to use as a comparable feature.

B. Proxy Family

The final pipeline evaluates a family of range-aware transformations:

- raw intensity I (baseline),
- direct range-aware proxy $I \cdot R$,
- aggressive proxy $I \cdot R^2$,
- log-stabilized proxy $\log(1 + I \cdot R)$,
- percentile-normalized $I \cdot R$,
- robust-scaled $I \cdot R$.

Stage 03 of the pipeline characterizes the behavior of each variant, and Stages 04–05 carry forward the most informative subset for global and semantic analysis.

C. Evaluation Metrics

Two main metrics are used. First, a Fisher-style multiclass separability score

$$S = \frac{\sigma_B^2}{\sigma_W^2 + \varepsilon}, \tag{5}$$

where σ_B^2 and σ_W^2 are the between-class and within-class variances of the signal, computed over the set of semantically valid points in a frame or window. Higher S indicates that classes are more separable along the signal axis. The per-frame gain is defined as

$$\Delta S = S_{\text{proxy}} - S_{\text{raw}}. \tag{6}$$

Second, for the short-window class-ordering analysis, Spearman rank correlation of class-mean signals is computed against the mean class profile of the window. Together, these give complementary views of separability and class-ordering stability.

III. Pipeline Overview

The final pipeline is organized into eight stages, implemented as a modular Python codebase under `src/` with reproducible entry-point scripts under `scripts/`. The eight stages are:

1) **Stage 01: Preliminary Work Recap** - Re-executes the midterm logic in one consolidated notebook to lock the baseline before any extension claims are made. Verifies dataset integrity, point-label alignment, raw and proxy signal behavior on a reference frame, and short-window semantic consistency.
2) **Stage 02: Multi-Window Sequence Setup** - Defines 15 reproducible evaluation windows over SemanticKITTI sequence 00: five short windows (length 10), five medium (length 30), and five long (length 100), spanning frames 000000–004540.
3) **Stage 03: Proxy Variant Builder** - Implements and characterizes the proxy family above on a representative window, including distribution behavior and stability checks.
4) **Stage 04: Multi-Window Global Signal Analysis** - Computes per-window descriptive statistics (mean, std, percentiles, range correlation) for each proxy across all 15 windows.
5) **Stage 05: Multi-Window Semantic Analysis** - Evaluates per-window Fisher separability for I, $I \cdot R$, and $\log(1 + I \cdot R)$, and computes per-window gains.
6) **Stage 06: Failure Case Analysis** - Classifies the 15 windows into failure, weak, and strong cases for each proxy, and compares signal distributions between weak and strong windows to identify the underlying failure mechanism.
7) **Stage 07: Video / GIF Asset Builder** - Builds the rendering primitives (BEV, first-person view, semantic overlays, signal comparison boards) under fixed normalization and deterministic subsampling, and produces frame-wise visual artifacts.
8) **Stage 08: Lightweight Downstream Study** - Tests whether $[I, I \cdot R]$ improves over $[I]$ in a simple logistic-regression classifier on a 30-frame medium window.

The companion artifact `final_seq.csv`, generated by `scripts/generate_gif.py`, logs frame-level raw and augmented separability over a 600-frame contiguous run and forms the basis of the headline quantitative results in Section V.

IV. Implementation and Visual Outputs

A. Modular Code Path

The implementation is split across the following modules:

- `src/io/`: frame loading and contiguous window sampling,
- `src/signals/`: pseudo-reflectivity and proxy variants,
- `src/metrics/`: Fisher separability and CSV logging,
- `src/semantics/`: class statistics and label mapping,
- `src/visualization/`: BEV, first-person view, and frame board renderers,
- `src/animation/`: temporal GIF writer.

The visualization layer enforces three rendering disciplines: fixed spatial limits, fixed normalization across frames (so brightness encodes actual signal value rather than per-frame rescaling), and deterministic subsampling. This is essential because per-frame normalization can mask poor signal behavior by producing visually pleasing but quantitatively misleading frames.

B. Single-Frame Signal Behavior

Figure 1 shows the distribution of raw intensity, $I \cdot R$, and $I \cdot R^2$ on a representative SemanticKITTI frame. The shapes differ substantially: raw intensity is concentrated below 0.5 with a sharp zero-spike, $I \cdot R$ retains a similar shape but stretches to a few tens, and $I \cdot R^2$ exhibits an extreme heavy tail extending to thousands. This is the empirical motivation for retaining $I \cdot R$ as the primary proxy and rejecting $I \cdot R^2$ as overcompensated.

Figure 2 presents the corresponding three-panel BEV comparison: semantic labels, raw intensity, and pseudo-reflectivity. The proxy is not a uniform rescaling of intensity. It changes which spatial regions are emphasized and exposes structure

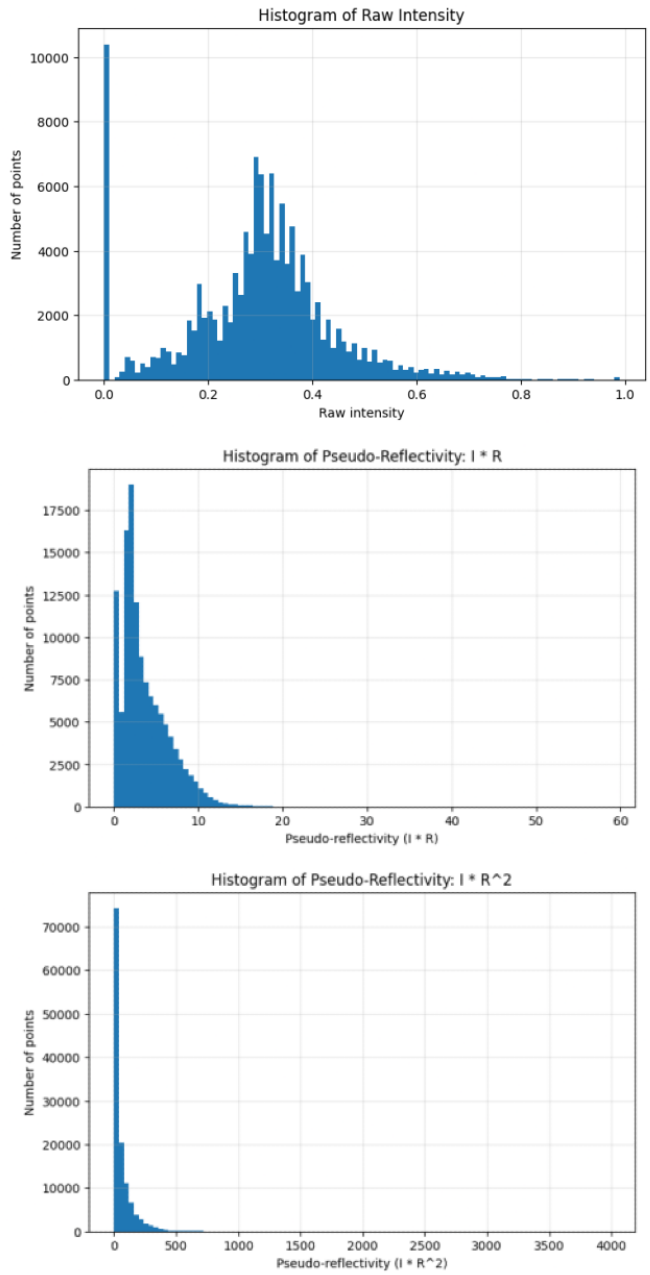

Fig. 1. Distribution of raw intensity, $I \cdot R$, and $I \cdot R^2$ on Frame 000000 of sequence 00. The $I \cdot R^2$ variant exhibits severe distribution explosion (x-axis extends to $\sim$4000), motivating its rejection in favor of $I \cdot R$ and $\log(1+I \cdot R)$.

that is suppressed under raw intensity, particularly at mid- and far-range points where intensity decay is most severe.

C. Class-Wise Mean Signal

Figure 3 shows class-wise mean values of raw intensity, range, and $I \cdot R$ for all valid semantic classes in the reference frame. Several class pairs that overlap heavily in raw intensity (for example road and sidewalk) become more separated under $I \cdot R$, which is the geometric reason the Fisher separability score improves.

TABLE I
CROSS-WINDOW STABILITY OF PROXY MEANS ACROSS 15 WINDOWS. CV IS COEFFICIENT OF VARIATION (STD/MEAN).

Proxy	Mean of means	Std of means	Var of means	CV
I	0.2937	0.0118	0.00014	0.0400
$I \cdot R$	3.3770	0.2167	0.04696	0.0642
$\log(1 + I \cdot R)$	1.2798	0.0499	0.00249	0.0390

D. Short-Window Class-Ordering Stability

Figure 4 reproduces the short-window semantic consistency analysis from the midterm baseline: Spearman rank correlation of class-mean signals against the mean class profile across a 10-frame motion window. Both raw intensity and pseudo-reflectivity preserve high class-ordering stability, with a mild advantage for the proxy on average, but with non-monotonic behavior across frames.

E. Temporal Separability Across Motion

Figure 5 shows per-frame Fisher separability for raw intensity and $I \cdot R$ across a 30-frame motion window. The two curves cross multiple times: pseudo-reflectivity is dominant in some frames and weaker in others. This is the first quantitative indicator that the proxy's advantage is scene-dependent, not uniform, and motivates the multi-window evaluation in Stage 05.

V. RESULTS

A. Multi-Window Global Signal Stability (Stage 04)

Table I summarizes per-window descriptive statistics aggregated across the 15 evaluation windows. Raw intensity is the most numerically stable but is also the lowest-variance signal. The direct proxy $I \cdot R$ is the most expressive but exhibits the highest cross-window variability. The log-stabilized variant $\log(1 + I \cdot R)$ achieves the best balance of expressiveness and cross-window stability, with a coefficient of variation comparable to raw intensity.

Figure 6 visualizes the per-window mean and standard deviation across the proxy family, including range correlation behavior.

B. Multi-Window Semantic Separability (Stage 05)

Figure 7 reports per-window separability gains, $\Delta S = S_{\text{proxy}} - S_{\text{raw}}$, for both $I \cdot R$ and $\log(1 + I \cdot R)$ across the 15 windows. The direct proxy $I \cdot R$ achieves strong positive gains in some windows (notably `short_2` and `short_4`) but produces clear negative dips in others (notably `short_3`). The log-stabilized proxy is consistently positive across nearly all windows and substantially reduces the magnitude of the negative dips. This is the strongest multi-window evidence in the project that proxy formulation matters as much as the underlying reflectivity-aware idea.

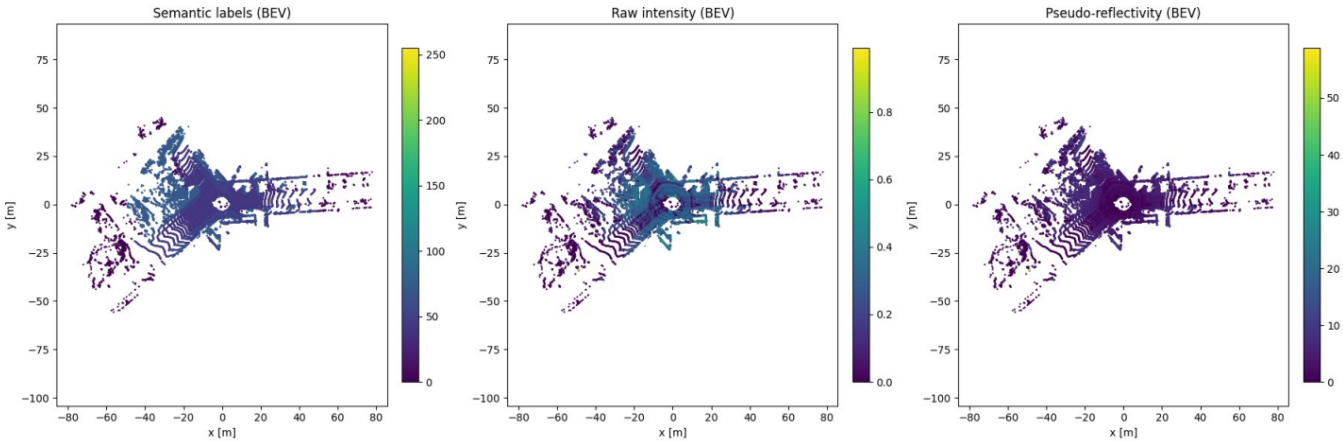

Fig. 2. Three-panel bird's-eye view of Frame 000000: semantic labels (left), raw intensity (center), pseudo-reflectivity $I \cdot R$ (right). Proxy emphasis differs spatially from raw intensity, particularly at mid- and far-range points.

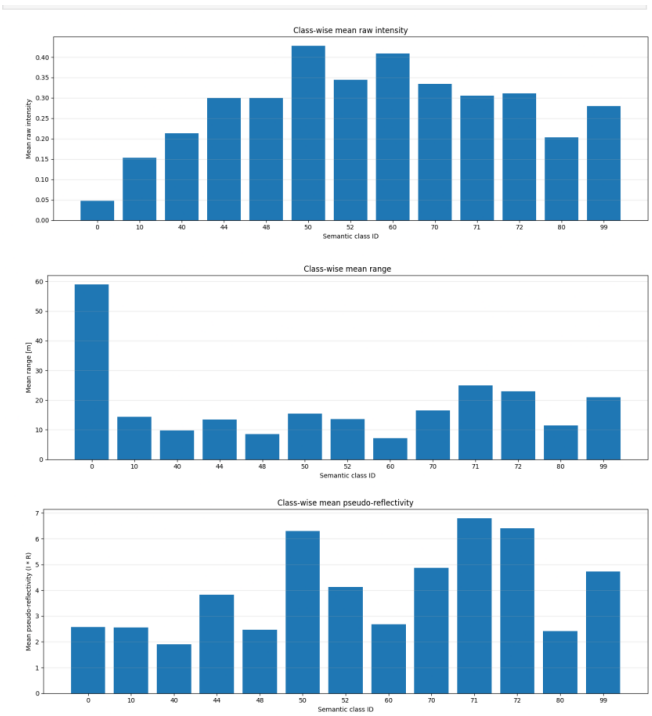

Fig. 3. Class-wise mean values for raw intensity, range, and pseudo-reflectivity $I \cdot R$. Range-aware scaling redistributes class means in a way that increases class separation along the signal axis, consistent with the observed Fisher separability gain.

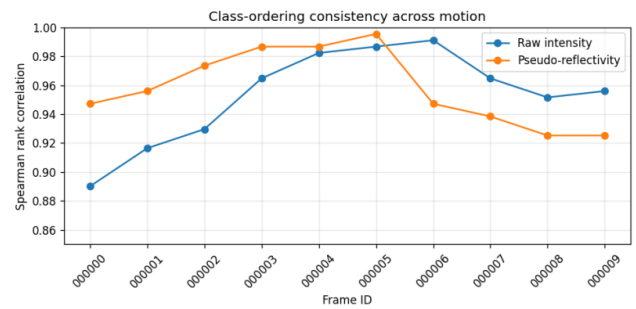

Fig. 4. Class-ordering consistency across a 10-frame motion window measured by Spearman rank correlation against the window-mean class profile. Both signals are highly stable; the proxy holds a mild but non-uniform advantage.

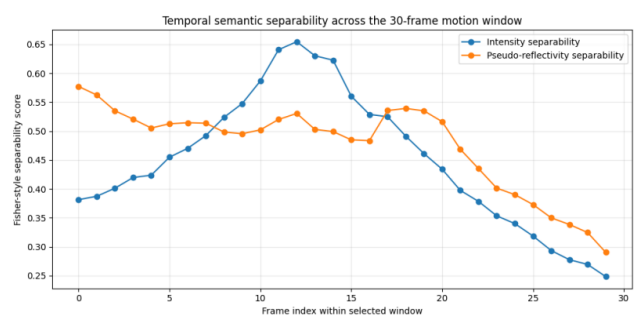

Fig. 5. Per-frame Fisher separability for raw intensity and pseudo-reflectivity across a 30-frame motion window. Curve crossings indicate that proxy advantage varies with scene composition rather than holding uniformly.

C. Failure-Case Analysis (Stage 06)

For the log-stabilized proxy, the 15-window classification yielded 0 failures, 1 weak case (`short_3`), and 14 strong cases. The weak/strong comparison was made between `short_3` and `short_2`. Figure 8 compares signal distributions between the weak and strong windows for raw intensity and $\log(1 + I \cdot R)$. The mechanism is not numerical collapse or missing data but rather class-wise signal compression: in the weak window, class-mean values are closer together along the signal axis, increasing inter-class overlap and reducing separability. In the strong window, class means are spread further apart and the proxy amplifies that separation.

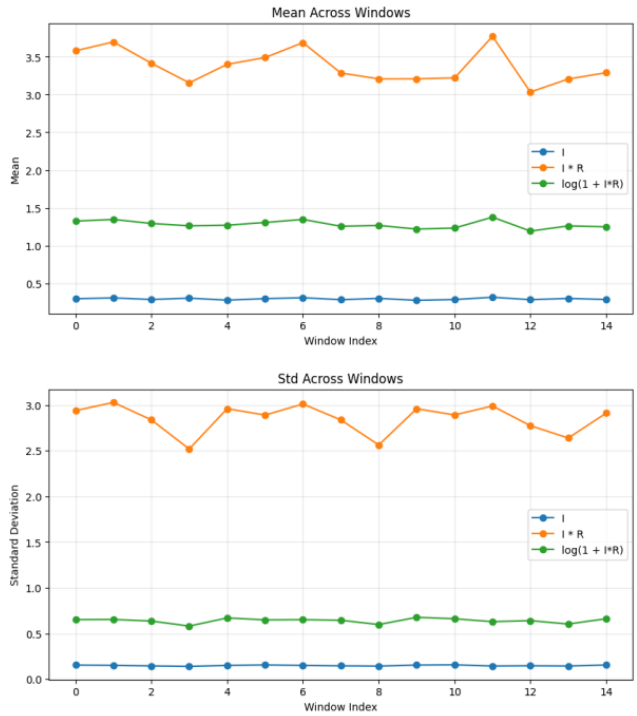

Fig. 6. Per-window mean and standard deviation for I, $I \cdot R$, and $\log(1+I \cdot R)$ across the 15 evaluation windows. The log-stabilized proxy maintains both expressiveness and low cross-window variability.

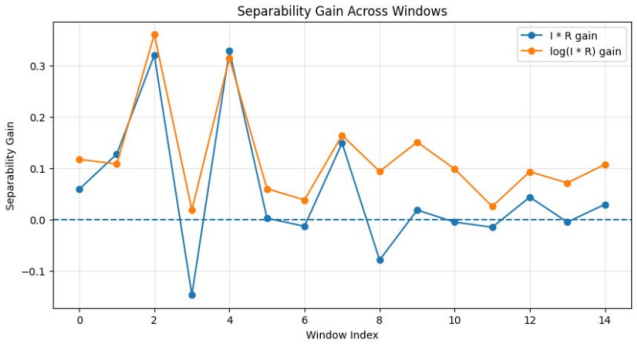

Fig. 7. Per-window separability gain ΔS for $I \cdot R$ (blue) and $\log(1 + I \cdot R)$ (orange) across the 15 evaluation windows. The log-stabilized variant is consistently positive while $I \cdot R$ exhibits clear failure modes, e.g. `short_3` and `short_8`.

D. Modular Pipeline Run: 600-Frame Artifact (Stage 07)

The artifact uses $I \cdot R$ as the augmented signal because per-frame visualization and separability logging benefit from a directly interpretable, range-aware quantity in native units, whereas the log-stabilized variant compresses dynamic range in a way that improves cross-window stability (Stage 05) but obscures per-frame magnitude. The two proxies serve complementary roles: $\log(1 + I \cdot R)$ as the multi-window recommendation, $I \cdot R$ as the per-frame artifact signal.

The modular pipeline runs over a 600-frame contiguous segment (sequence 00, frames 000211–000810) and produces

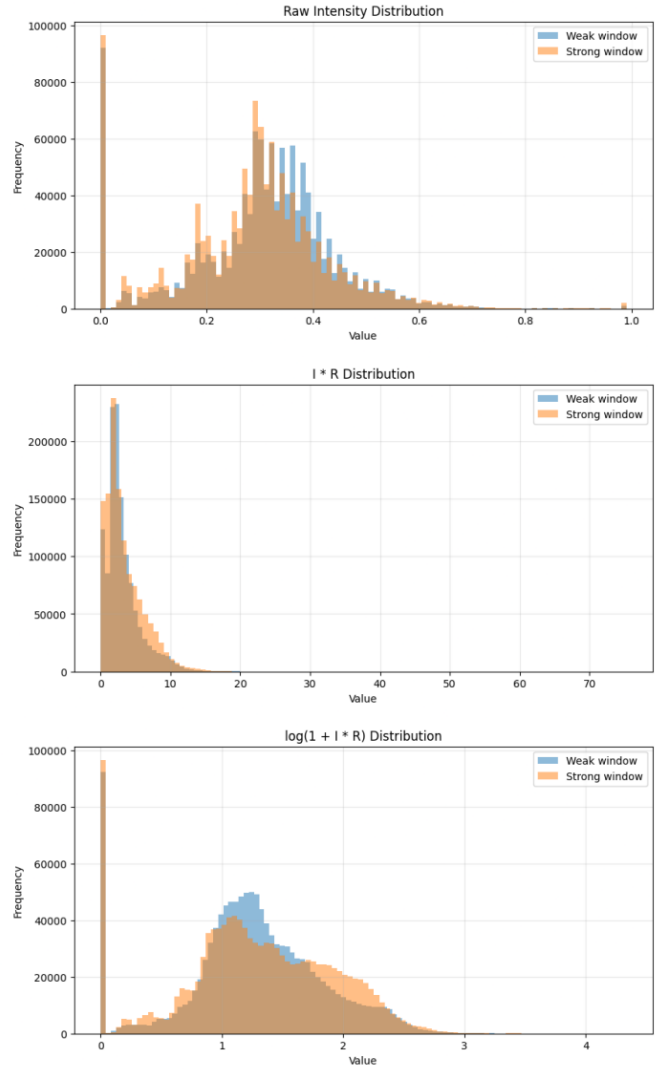

Fig. 8. Signal distributions in the weak window (`short_3`) versus strong window (`short_2`) for raw intensity (top), $I \cdot R$ (middle), and $\log(1 + I \cdot R)$ (bottom). The failure mechanism is class-mean compression rather than numerical collapse; the proxy formulations show greater separation between weak and strong windows than raw intensity does.

`final_seq.csv`, logging per-frame raw separability S_{raw}, augmented separability S_{aug}, and delta $\Delta = S_{\text{aug}} - S_{\text{raw}}$. Aggregate statistics are summarized in Table II.

The mean delta is positive, but two cautions apply. First, 200 of 600 frames show negative gain, and the worst-frame loss is comparable in magnitude to the best-frame gain, so the average is the result of a heavy mixture rather than a uniform improvement. Second, consecutive frames in a contiguous run are not independent samples: per-frame deltas exhibit very high temporal autocorrelation (lag-1 autocorrelation ≈ 0.98), so the 600 frames represent a substantially smaller number of effectively independent observations. Confidence intervals computed under a naive independence assumption therefore overstate the precision of the mean estimate. The headline number is reported with these limitations made explicit rather

TABLE II
AGGREGATE STATISTICS FROM `FINAL_SEQ.CSV` (600 FRAMES, SEQUENCE 00).

Metric	Value
Mean raw separability	0.3018
Mean augmented separability	0.3455
Mean delta (Δ)	$+0.0436$
Relative gain (vs raw mean)	$+14.45\%$
Positive-gain frames	400 / 600
Negative-gain frames	200 / 600
Best frame (delta)	`000743` ($+0.4613$)
Worst frame (delta)	`000355` (-0.4267)

TABLE III
LIGHTWEIGHT DOWNSTREAM CLASSIFIER RESULTS (LOGISTIC REGRESSION).

Configuration	Accuracy
$[I]$ only	0.3779
$[I, I \cdot R]$	0.4090
Absolute gain	$+0.0311$

than smoothed over.

Figure 9 shows a representative frame from the rendered artifact, combining BEV, first person view, and a top-class summary panel under fixed normalization across frames.

E. Lightweight Downstream Study (Stage 08)

The direct proxy $I \cdot R$ is used here because logistic regression on raw scalar features benefits from preserving native dynamic range; the log-stabilized variant, while preferable for cross-window aggregate comparison (Stage 05), compresses the very signal range that a linear classifier exploits. The two proxies remain complementary across the pipeline.

Stage 08 tests whether the proxy provides feature-level value in a simple classifier. Using a single 30-frame medium window (`medium_0`) with 3,621,402 valid points after class filtering, the classifier is trained on a random subsample of 200,000 points (80/20 train-test split) using logistic regression with `StandardScaler` preprocessing. Two feature configurations are compared: $[I]$ and $[I, I \cdot R]$.

The $+3.11$ percentage-point overall gain is concentrated in a small subset of classes:

- Road (40): $+11.6\%$ (0.7023 $\rightarrow$ 0.8181),
- Sidewalk (48): $+7.6\%$ (0.0000 $\rightarrow$ 0.0765),
- Class 81 (other-object): $+5.2\%$,
- Building (50): $+0.4\%$,
- Vegetation (70): -3.6%.

Two important caveats apply to the downstream result. First, several minority classes have zero baseline and zero augmented accuracy, indicating that logistic regression on one or two scalar features is too weak to discriminate them at all. The headline gain is therefore driven almost entirely by the dominant structured classes (road, sidewalk, building). Second, the result is reported as an exploratory feature-level signal check, not as a deployment-grade segmentation claim. Demonstrating practical downstream utility in a realistic system would require a more expressive model and a richer feature set that contextualizes pseudo-reflectivity alongside geometry and local descriptors.

VI. DISCUSSION

The results across the eight pipeline stages converge on a clear finding: a lightweight, range-aware transformation of LiDAR intensity provides measurable, scene-dependent improvements in semantic interpretability. Three observations make this concrete.

A. Proxy Design Matters as Much as the Idea

The midterm work focused on $I \cdot R$ and demonstrated feasibility. Stage 05 shows that the choice of proxy formulation has a first-order effect on whether the reflectivity-aware idea generalizes. Both $I \cdot R$ and $\log(1 + I \cdot R)$ encode the same underlying insight, but exhibit substantially different stability profiles across the 15 windows. The log-stabilized variant produces consistent positive gains; the direct variant exhibits explicit failure modes. Promoting $\log(1+I \cdot R)$ from analysis-side variant to primary recommendation is one concrete output of the final stage.

B. Failure Cases Are Diagnostic, Not Anomalous

Stage 06 identifies one weak window for the log-stabilized proxy and traces its failure mechanism to class-mean compression rather than numerical collapse. This predicts that the proxy will be most useful in scenes where class-mean signal values are spread apart, such as urban driving scenes with distinct road, sidewalk, building, and vegetation surfaces. It will be least useful where class signals already overlap heavily, since range-aware scaling cannot create separation that is not latent in the raw signal.

C. Class-Wise Behavior Aligns With Geometry

The downstream class-wise results in Stage 08 are consistent with the failure-case analysis. The classes that benefit most (road, sidewalk) are large, structured, geometrically consistent surfaces at predictable ranges. The class that loses (vegetation) is geometrically heterogeneous, with high intra-class variability driven by leaves, branches, and incidence-angle variation. Range-aware scaling amplifies this variability rather than reducing it, increasing class overlap. The proxy is therefore not a uniform improvement but a structured, class-dependent reweighting.

D. Scope and Boundaries

The contribution is deliberately scoped to signal-level analysis rather than calibrated reflectivity recovery, keeping the method lightweight and interpretable; material identification sits as a natural extension of this foundation. Sensor-specific calibration effects such as gain modeling, near-range artifacts, and multi-return handling are not modeled here, leaving room for richer calibration to refine the proxy further. The 600-frame run draws from a single contiguous segment of one sequence, providing a clean reference that future runs over

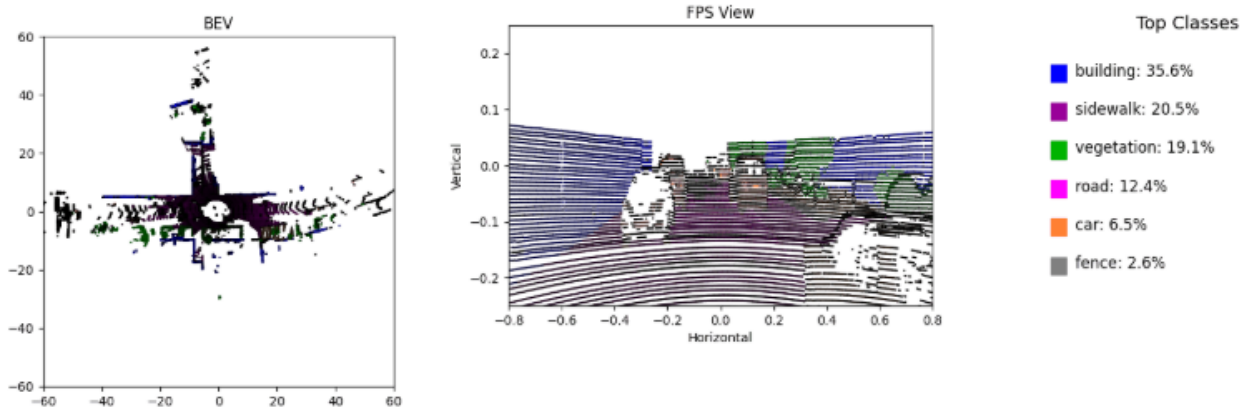

Fig. 9. Representative frame from the modular pipeline artifact: BEV (left), forward-perspective view (center), and top-class statistics panel (right). All panels use fixed normalization and deterministic subsampling so brightness reflects actual signal magnitude rather than per-frame rescaling.

multiple sampled windows can build on. The high temporal autocorrelation of per-frame deltas also clarifies the right inferential framing: window-level resampling and block-bootstrap will tighten the statistical characterization in subsequent evaluations. The downstream classifier is intentionally minimal, a controlled probe rather than a deployment system, so that any gain reflects feature-level signal rather than model capacity.

E. Potential Extensions

The framework established here opens several promising directions. Window-level confidence intervals would strengthen the statistical characterization of per-frame deltas. Integrating $\log(1 + I \cdot R)$ as a feature channel in a learned segmentation model would test the proxy's value beyond logistic regression. Estimating incidence angle from local surface normals would move $\hat{\rho} = I \cdot R$ closer to a physically motivated correction. A class-conditioned failure-case analysis would further sharpen the scene-dependence story already surfaced by this work.

VII. Conclusion

Building on a midterm feasibility study, this work develops a structured, multi-stage evaluation pipeline for reflectivity-aware LiDAR signal augmentation. The analysis establishes that $\hat{\rho} = I \cdot R$ is a meaningful signal transformation whose benefits track interpretable scene structure, and that $\log(1 + I \cdot R)$ delivers more consistent improvements across temporal windows. Frame-level Fisher separability improves across a 600-frame contiguous run, and a lightweight downstream classifier surfaces clear feature-level gains on structured surface classes. The result is a disciplined characterization of a simple, physically motivated idea: lightweight range-aware augmentation reliably enhances LiDAR signal interpretability under explicit conditions, standing as a controlled feature transformation that future systems can build on with confidence.

VIII. Project Link

https://github.com/aymisxx/reflect-aug-seg

Acknowledgment

This project was completed as part of **SES 598: Space Robotics & AI** at Arizona State University, under the guidance of Prof. Jnaneshwar Das. The course is affiliated with the **Distributed Robotic Exploration and Mapping Systems (DREAMS) Laboratory**. The overall project framing, evaluation discipline, and robotics context were shaped by the course environment and by broader research themes in robotic perception, exploration, and autonomous systems associated with that academic setting. This work represents an independent implementation and analysis carried out within that context.

Statement on Use of Generative AI

Generative AI tools (ChatGPT by OpenAI and Claude by Anthropic) were used to assist with drafting, editing, code organization, and debugging support throughout this work. All technical decisions, implementations, and results were critically reviewed, validated, and integrated by the author, who assumes full responsibility for the final submission.

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[1] K. Viswanath, P. Jiang, and S. Saripalli, "Reflectivity Is All You Need!: Advancing LiDAR Semantic Segmentation," arXiv preprint arXiv:2403.13188, 2024. [Online]. Available: https://arxiv.org/abs/2403.13188

[2] J. Behley, M. Garbade, A. Milioto, J. Quenzel, S. Behnke, C. Stachniss, and J. Gall, "SemanticKITTI: A Dataset for Semantic Scene Understanding of LiDAR Sequences," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 9297–9307. [Online]. Available: https://openaccess.thecvf.com/content_ICCV_2019/papers/Behley_SemanticKITTI_A_Dataset_for_Semantic_Scene_Understanding_of_LiDAR_Sequences_ICCV_2019_paper.pdf